

Muhammed Can Küçükaslan

Senior Backend Software Engineer

E-mail muhammedcankucukaslan@gmail.com muhammed@kucukaslan.com.tr
GitHub <https://github.com/kucukaslan> **Website** <https://kucukaslan.com.tr/>
LinkedIn <https://linkedin.com/in/kucukaslan/>

Education

2019 – Jan. 2024 **Bilkent University**, B.S. in **Computer Engineering** & Mathematics (*Double Major*)
Bilkent University, Faculty of Engineering, Ankara, Turkey.

- **GPA: 3.66/4.00 (CS), 3.65/4.00 (Math) | Magna Cum Laude (Top 10%)**
- Comprehensive Scholarship (100% Tuition Waiver)

Work Experience

Aug. 2026 - now **Getir Turkish Decacorn**
Senior Backend Engineer I

- Investigated service memory usages and found Kubernetes requests and limits had been sized around reclaimable filesystem cache. Right-sized relevant deployments; relevant EC2 spend fell by \$30 per day, roughly \$11k annualized.
- Improved request distribution across Kubernetes pods by tracing how connection pooling fails when used with ClusterIP. After adding a proper L7 routing proxy, most client timeouts were eliminated and tail latency was shortened: e.g. the share of requests completed under 20 ms rose from 70% to 84%.

Feb. 2025 - **Getir Turkish Decacorn**
July 2026 (1 year 6 months) **Backend Engineer II (Mid)**

- Led IoT integration for car tracking system, building device-specific protocols on scalable Java & Go microservices.
- Optimized and maintained map and distance estimation services handling 350k RPM.
- Customized internal OSRM deployments to *incorporate historical delivery data and GPS traces* using AirFlow, PostgreSQL, and Redis.
- *Proactively* audited Kubernetes resource limits and HPA policies, achieving a **\$20,000 reduction** in annual **infrastructure costs**.
- Collaborated on migrating an internal vehicle routing service from **Python to C++**, *doubling performance* and **saving \$10,000+ in monthly recurring costs**.
- Deployed infrastructure (Kafka, MongoDB, K8s, EC2) with strict ACL settings using **Terraform** and Cloud **CDKs**.
- Migrated CI/CD pipelines from Bitbucket to GitHub Actions.
- Mentored 2+ interns/junior engineers and authored company-wide guides for MongoDB upgrades to improve cross-team knowledge sharing.

Jan. 2024 - **Backend Engineer I - Java (Junior)**
Jan. 2025 (1 year)

- Design and implementation of courier rewarding (& gamification) service in **Go** with **Go-kit** with RESTful microservice architecture using Kafka and Flink.
- Maintained **Java/Spring** microservices, ensuring reliability through comprehensive testing (JUnit, WireMock) and distributed scheduling (ShedLock).
- Used Apache Kafka, Redis, MongoDB, **AWS** Lambda, Docker, Kubernetes, Bitbucket Pipelines.
- Optimized, in a joint effort, MongoDB usage to dramatically improve certain query performances, database migrations/consolidations, and **reduce monthly database costs** by \$2000+.

- July 2023 - **OTO Global (Remote) Backend Engineering** Intern & Part Time
- Nov. 2023 (5 months) • Developed backend systems using **Java & Spring**, deployed on Google Cloud Run (GCR) & Compute Engine with **Docker** images built with **Jib**.
- Monitored system health using GCR Metrics and Logging to identify anomalies, while managing data with MySQL, Redis, and Pub/Sub.
- June 2022 - **Peak Games (1st Turkish Unicorn) Big Data Platform Engineering** Intern
- Sep. 2022 (4 months) I worked on the development of [s5cmd](#) written in **Go**. [s5cmd](#) is a command line application used for cloud storage operations, particularly for AWS S3.
- Implement extensive shell autocompletion for zsh, bash and pwsh [#500 · peak/s5cmd \(github.com\)](#)
 - Implement versioning support for S3 compatible file systems [#475 · peak/s5cmd](#)
 - Refactor sync command to handle sync of directories with millions of files under limited RAM [#483 · peak/s5cmd \(github.com\)](#)
 - Identified and fixed a subtle path manipulation bug [#456](#) and more at [Pull requests · peak/s5cmd \(github.com\)](#).

Select Public Work

- Investigated uneven pod load caused by connection reuse, wrote about [Kubernetes connection-level load balancing](#) and built an [interactive HTTP routing simulator](#) to demonstrate the behavior.
- Diagnosed a long-standing geographic extraction bug in osmium-tool, produced a focused reproduction, and [contributed the upstream fix](#). I [documented the investigation](#).
- Published a [FastAPI and Uvicorn demo](#) showing production server configuration, Kubernetes deployment, and response compression, with a [technical write-up](#) explaining the tradeoffs.
- Built [PoCSequentiality](#), a Go prototype for downloading large S3 objects with sequential disk writes to reduce random I/O

Skills

Programming languages	Python: FastAPI, NumPy, Pandas, OR-Tools Go: Go-kit, urfave/cli, Goroutine patterns Java: Spring Boot, OOP Other: C++, SQL, NoSQL (MongoQL, Redis)
Various development tools and environments	Git (GitHub, BitBucket and GitKraken) MongoDB, PostgreSQL, ClickHouse, MariaDB, Redshift, Athena CSV, JSON, Protocol Buffers, Parquet Apache Kafka , Google Pub/Sub Docker, Kubernetes CI/CD (GitHub Actions, BitBucket) pipelines AWS (EC2/S3), Google Cloud (Run, Storage) [GCP]

Other Information

Achievements & Rewards

- Bilkent University Comprehensive **Scholarship** Student (includes *100% tuition waiver* and stipends), 2019-2024.
- Ranked **568th** in “YKS” among over **2 million candidates**, in “Sayısal” (lit. Quantitative), **123rd** in “Sözel” (lit. Verbal), **475th** in English.
- Bilkent University *High Honor* Certificates in **Computer Eng.** (7x), in **Mathematics** (6x).
- Bilkent University *Honor* Certificates in **Computer Eng.** (2x), in **Mathematics** (once).

Certificates

- **Go: The Complete Developer's Guide (Golang)** by *Stephen Grider*
Certificate no: UC-c48f8569-b251-45f8-a057-3d491de174ca